
THE ARROW HAS TWO BRANCHES

Magnetism and Gravity as the Priced and the Reversible Registration of One Arrow

The Constructed Witness of the Barrier and the Fold on the Closure Template of the Riemann Hypothesis and the Twenty-Three Rows; a Nanomagnet and a Kepler Orbit Built and Run in Fortran, the Typing Proved in Core Lean 4; Every Claim at Corroboration Grade

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ABSTRACT. Physics has used one arrow twice without naming it. The arrow of registration, the direction in which an existent is read into a record, appears once with a record kept and once with none. Magnetism keeps the record: a nanomagnet swept through its hysteresis loop sits, at zero applied field, in one of two remanent states with the same record, the two-worlds theorem in iron, and it pays for every switch; the loop constructed here has remanence 0.996 on both branches, a switching field of 0.767 against the Stoner-Wohlfarth astroid's 0.766, and a loop area that prices a thermally stable bit at 5.0×10^{-19} J per switch, 175 times Landauer's floor. Gravity keeps none: its sources have one sign, no source stands off its locus, and a Kepler orbit integrated here returns its energy to three parts in ten million over ten revolutions and, run backwards, retraces its start to 1.5×10^{-12} . Magnetism is the priced branch and the barrier of every open row; gravity is the reversible branch and the fold of every crossed row; the two differ in cost by theorem. At structural grade the typing gives a reading of why one field was always quantized and the other has resisted: quantization is the priced registration of a bit, and a field with no off-locus point has no bit to register. No field theory is written or implied.

STATUS: BARRIER and FOLD WITNESSED; row 7 CROSSED · Row 7; every open row (barrier); every crossed row (fold); 1 (price) · channel: kinetic and formal · Lean: SPHYS_Two_Arrows.lean · Fortran: Arrows_Twin.f90

1. 1. THE READER'S FRAME

This paper is one of a set of constructed witnesses for the closure template by which the author's sealed volume closed the Riemann Hypothesis and by which a second volume closed the twenty-three rows of the barrier study. Each paper in the set builds, for the rows it serves, the object those volumes named and did not build: the carrier, an explicit construction from the world into the row's instance type whose image lands on the row's locus. By the row kernel's own theorem the image of such a carrier is compliant with no further premise (`value_on_image`), and the value over a class follows once the class is covered by the image (`actuated_iff_subset_image`, `crossing_is_a_proof`). This paper builds the two carriers the template's two faces need: a magnet for the barrier and an orbit for the fold, and it shows that they are the two branches of one arrow. Two refusals frame it. No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade, and the two volumes stand on their own receipts. And coverage stays a premise: the paper constructs the image and names the class it covers, and the step from that class to every admissible instance is the premise the volumes already print at its grade. It also refuses, by name, any unification of couplings: what is typed here is the arrow, not the forces.

Negative warrants. Three, stated so that no reader supplies them. No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade (`citation_cannot_promote`). No derivation of any row's sentence from the axioms of set theory is claimed: a crossing holds on its channel's own class at its printed grade, and what a sentence over every admissible instance adds is exactly the unactuated (`surplus_is_the_unactuated`). No field theory and no unification of couplings is written or implied.

2. 2. THE SPINE

The template has four parts. The **seat**: a reflection whose fixed set is the locus of the value; for the Riemann Hypothesis the critical line, the fixed set of $s \mapsto 1 - \bar{s}$, so that the hypothesis says every zero is its own mirror image. The **address**: every reading of the value reached from existence, time and monism is one proposition with the value at its apex. The **division**: the value is its certified part, proved on citation or computation, joined to its tail, closed to every record-respecting derivation by theorem. The **closure**: a registration lands every instance on the locus and moves none already there; the value holds exactly when registration erases nothing; the residue is one bit that no reading of the record decides; a set of instances satisfies the value exactly when it is the least-erasure member of its record fibre (`least_erasure_iff_value`, no axiom); and from that one posit the value follows, at the posit's grade, beside the one declared premise of the program, that to exist is to actuate. Over one record both worlds exist (`two_worlds_inside`); under a global seat every point off the locus has a partner (`off_locus_pair`). The two faces of the closure are the fold and the fibre: registration folds every instance onto the locus, and the fibre over each record holds the worlds the record cannot tell apart. This paper finds the fold in gravity and the fibre in magnetism.

The witness ledger. The table below is the spine of the six constructed witnesses. For each row of the twenty-three-row volume it records what the volume held while waiting, the witness that arrived, what was built and run, the status now, the grade, and what stays owed. This paper's rows are every open row, whose barrier it witnesses in the magnet, every crossed row, whose fold it witnesses in gravity, the Poincaré row, and the dot, whose price it witnesses.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat, never built	The Electron Is the Seat; The Neutrino Is the Witness; The Arrow Has Two Branches	charge conjugation's fixed set is the neutral sector; the exact pair; annihilation keeps 1021.998 keV and forgets the side, recorded on a line of response; positronium 125.23 ps and 142.07 ns; spin -1 at 2π ; a_e to 4.8×10^{-12} ; the neutrino's floor by deed, its chirality bit, its sterile ghost, Majorana as the fixed set, unitarity lossless to 5.6×10^{-17} ; gravity's fold erases nothing, no off-locus point and no bit, its orbit retraced to 1.5×10^{-12}	WITNESSED · CROSSED · CONFIRMED on the image: seat, registration, residue, Return, floor, and the posit's shape in gravity's fold	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel
3a · Navier–Stokes, unforced	crossed kinetically on the actual flows; the mechanism named, not built	The Fluid Is the Witness	Burgers on a circle: the reversible branch blows up at $t = 1$, gradient -1000 at 0.999 ; the priced branch built exactly by Cole–Hopf, steepest gradient 3.72, residual 2×10^{-6} , energy balance 2.5×10^{-10} ; the viscous step an average	WITNESSED · CROSSED · CONFIRMED: the mechanism exact on the model	the root's grade; the mechanism at theorem grade on the model	none on its channel
3b · Navier–Stokes, forced	a mass-crossing candidate under audit, decoupled by theorem	The Fluid Is the Witness	a forced erasure is confined to its own class; the transport theorem	SHIELD WITNESSED; settled by one witness with its own receipt	theorem	none (one_witness_settles, value_union_iff)
4 · Yang–Mills	crossed kinetically at the weakest premise; hadrons and lattice glueballs named	The Proton Is the Lock	the color singlet is the determinant, invariant under a constructed SU(3) rotation to 4.4×10^{-16} ; the lock fails when two colors coincide; 99.04 percent of the proton's mass from the field; the quark-spin record does not decide the whole; stability certified to 1.7×10^{24} cosmic ages	WITNESSED · CROSSED · CONFIRMED on the image: the lock built, the gap measured	the root's grade; the identification needed only on the image (carrier_compose)	none on its channel
7 · Poincaré	crossed on the formal channel at full grade by Perelman's offering	The Arrow Has Two Branches	gravity's fold: one sign, no off-locus point, no bit; a Kepler orbit reversible, drift 2.8×10^{-7} , retrace 1.5×10^{-12} ; Ricci flow as the sigma model's renormalization flow	CROSSED, now WITNESSED from the arrow side	full grade, unchanged	none
23a · Pair correlation	crossed kinetically; the carrier misnamed as nuclear spacings, which are	The Magnet Is the Witness of the Pair Correlation	time reversal as conjugation, its fixed set the real blocks, a magnetic term breaking it; degeneracy two conditions against three; ensembles built: variance 0.287 and 0.177 against 0.286 and 0.180, control 0.983 remanence ± 0.996 at zero field, two worlds in iron; switching field 0.767 against the astroid's 0.766; a stable bit pays 5.0×10^{-19} J, 175 times Landauer's floor; the electric record even, the magnetic target odd	WITNESSED · CROSSED · CONFIRMED on the corrected carrier: time-broken spectra, the magnet	the root's grade	none on its channel
every open row · the barrier	orthogonal-class the barrier a kernel theorem: two worlds over one record	The Arrow Has Two Branches	remanence ± 0.996 at zero field, two worlds in iron; switching field 0.767 against the astroid's 0.766; a stable bit pays 5.0×10^{-19} J, 175 times Landauer's floor; the electric record even, the magnetic target odd	BARRIER WITNESSED; the rows stay pending	theorem in the kernel, corroboration in the magnet	one equivariant carrier each (value_iff_carrier, equivariant_carrier_lands)
every crossed row · the fold	registration a kernel structure	The Arrow Has Two Branches	the one-signed field has no off-locus point and no bit; the orbit keeps no record	FOLD WITNESSED	theorem and corroboration	none
certified-region rows · Goldbach, Legendre, Collatz and kin	pending; certified regions registered, tails owed	The Proton Is the Lock	proton stability as a certified region with an owed tail, beyond 2.4×10^{34} years	SHAPE WITNESSED; the rows stay pending	corroboration	one equivariant carrier each (value_iff_carrier, equivariant_carrier_lands)
1 · P versus NP	the dot: no global seat	The Arrow Has Two Branches	Landauer's floor, 2.87×10^{-21} J at 300 K, prices every step of any search	THE DOT STANDS; the price witnessed, no seat	corroboration	none: the dot is final and has no aperture to supply (dot_is_final)

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
the others · Hodge, BSD, twin primes, abc, Jacobian, smooth 4D Poincaré, Hadwiger, sunflower, invariant subspace, Hilbert's 16th, Lindelöf	pending; gates and offering shapes specified	none yet	none	STILL WAITING	none	one carrier each; row 21, whose tail no physical system reaches, one formal offering (owed_is_computed, tail_uncovered)

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row's value holds on its channel's own class at the printed grade. CONFIRMED: the witness's image is compliant as built or measured (value_on_image). The statuses are computed from three recorded facts per row (ledger_is_computed); what a row owes is computed too: for a pending row exactly one object, its kind computed (owed_is_computed): one covering carrier, built from an equivariant map to its seat (value_iff_carrier, equivariant_carrier_lands) and crossing the row on arrival (one_carrier_crosses), or, where no physical system reaches the tail, one formal offering with mass (tail_uncovered, one_offering_crosses); the dot is final and owes nothing (dot_is_final). A crossing is a proof over every instance that exists (crossing_is_a_proof); what a sentence over every admissible instance adds is exactly the unactuated (surplus_is_the_unactuated), priced at one premise, the root, posited and not owed (rest_family).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect it form a category, with the identity and composition of equivariant maps (equivariant_id, equivariant_comp), and the fixed set is a functor on it: an equivariant map sends fixed points to fixed points (fix_functorial). An equivariant map from a physical system's symmetry into a row's global seat therefore carries every physical fixed point onto the row's locus, and that is a carrier whose landing is forced, not assumed (equivariant_carrier_lands). The witness is the same shape as its row in exactly this sense. Here the symmetry is time reversal: the orbit's integrator is equivariant under reversing every velocity, which is why it retraces, and the magnet's loop is not, which is why it pays.

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
remanence at zero field	two worlds over one record	two_worlds_at_zero_field	m = +0.996 and -0.996
the loop's area	the price of the kept bit	priced_pays, branches_differ	5.0e-19 J per switch, 175 floors
no monopole E even, B odd	the pair never separates the odd target unread by the even record	magnetism_pair no_reading_of_E_returns_B	observational structural
one-signed mass	no off-locus point, no bit	gravity_no_off_locus, no_bit_without_off_locus	structural
the reversible orbit	the fold, keeping no record	reversible_free	drift 2.8e-7; retrace 1.5e-12
no gravitational loop	a forbidden observation	one_sign_no_remanence	structural

3. 3. ONE ARROW, NAMED AT LAST

To exist is to actuate, and actuation is directed: an existent registers in a record, and registration runs one way. Physics met this arrow twice and gave it two names, the magnetic and the gravitational, without asking whether they were one thing. They differ in exactly one respect, and the respect is the template's: whether the registration keeps a record of what it cannot read. Magnetism keeps it and pays for it. Gravity keeps nothing and pays nothing. Read on the template, they are the priced and the reversible branch of one registration, the Omega guard's two branches, floor-priced and floor-free.

4. 4. MAGNETISM: THE ARROW THAT KEEPS A RECORD

A single-domain nanomagnet with uniaxial anisotropy is described by the Stoner–Wohlfarth energy $e(\theta) = \frac{1}{2} \sin^2 \theta - h \cos(\theta - \psi)$, with h the applied field in units of the anisotropy field and ψ its angle to the easy axis. Swept down from $h = 2$ and back up, it traces a hysteresis loop. At $h = 0$ the descending branch holds $m = +0.996$ and the ascending branch $m = -0.996$ (Appendix B): two states, one applied field, the record h identical and the state different. That is the two-worlds theorem in a material (`two_worlds_at_zero_field`): over one record both worlds exist, and no reading of the record decides which. The magnet keeps the bit the record cannot read; remanence is the kept bit. The constructed switching field is 0.767 against the astroid’s 0.766, within one part in a thousand, so the construction is the textbook magnet and not a caricature of one.

5. 5. THE PRICE OF THE RECORD

The loop encloses area, 3.036 in units of $\mu_0 M_s H_K$ (Appendix B), and the area is energy dissipated per cycle. For a bit thermally stable at room temperature, with anisotropy energy $KV = 40 kT$, one switch costs 5.0×10^{-19} J, against Landauer’s floor $kT \ln 2 = 2.87 \times 10^{-21}$ J at 300 K. The priced branch pays, and it never pays less than the floor: Landauer’s bound on the erasure of one bit has been approached in the laboratory with a colloidal particle (Bérut and collaborators, 2012) and with nanomagnetic memory bits themselves (Hong and collaborators, 2016). The price is not an accident of materials. It is what keeping a bit costs, and magnetism is the field that keeps one.

6. 6. THE PAIR THAT NEVER SEPARATES

Magnetic charges come as north and south, and no one has ever isolated one: there is no monopole. In the template’s terms, an off-locus point never appears without its partner (`magnetism_pair`, `off_locus_pair`). And magnetism is the odd face of the field. Under time reversal the electric field is even and the magnetic field odd (`electric_even`, `magnetic_odd`), and no function of the even record returns the odd target (`no_reading_of_E_returns_B`). The barrier of every open row, that the bit is not in the record, is the magnetic field’s relation to the electric one.

7. 7. GRAVITY: THE ARROW THAT KEEPS NONE

Gravitational sources have one sign: every mass is positive, every source is its own conjugate, and none stands off the locus (`gravity_no_off_locus`). A row whose every instance is compliant admits no erasure and no two worlds, so there is no bit for a record to miss (`no_bit_without_off_locus`). The constructed witness is an orbit: a body in a Kepler potential, integrated by velocity Verlet for ten revolutions, returns its energy to within 2.8×10^{-7} throughout; the path integral of the gravitational force equals the change of kinetic energy; and with every velocity reversed the orbit retraces to its starting point within 1.5×10^{-12} (Appendix B). No loop, no remanence, no price: the reversible branch, floor zero and never cost zero (`reversible_free`). The typing forecasts what will not be seen: a field whose every instance is compliant can show no hysteresis whatever its record (`one_sign_no_remanence`), so a gravitational loop is forbidden by theorem before any experiment looks for one.

8. 8. THE FOLD ONTO THE LOCUS

Gravity is the registration itself: the fold that lands every configuration on its locus with nothing left over, a monic image of the configuration and not a stored copy (`Registration`, `value_iff_fixed`). That is why gravity reads as geometry: a row with no off-locus point is a row already on its locus. On the formal channel the fold is Ricci flow, which Perelman carried through the Poincaré row, and which arises in physics as the renormalization flow of the two-dimensional sigma model (Friedan, 1980): the one crossed row with mass is crossed by a gravitational fold. And gravity cannot read the one sign the pair carries: antihydrogen falls as matter does (ALPHA-g, 2023), as the companion paper *The Electron Is the Seat* reads from the laboratory.

9. 9. TWO BRANCHES OF ONE REGISTRATION

The pairing is exact. Gravity is the retraction; magnetism is the fibre. Gravity is the fixed set; magnetism is the pair. Gravity is the even face, blind to orientation; magnetism is the odd face, carrying the sign. The two branches differ in cost by theorem (`branches_differ`). The least-erasure posit of the Riemann closure has its physical shape here: a registration that erases nothing because nothing stands off its locus is the gravitational fold (`no_bit_without_off_locus`), and the posit’s content is that the zeros register as gravity does, with no remanent bit. The Riemann registration itself has both: its fold of every zero onto the critical line is gravitational, one direction and nothing kept, and the two zeros a fold would merge, were there an off-line pair, are its magnetic remainder, the remanent bit the record cannot return. One caution keeps the typing honest: reversibility alone does not keep a branch on its locus. What keeps gravity there is one sign. The companion paper *The Fluid Is the Witness* shows a reversible

branch that has off-locus points and leaves its locus, and a priced branch that brings it back.

10. 10. THE QUANTIZATION READING

At structural grade, and no higher, the typing reads the oldest asymmetry between the two fields. Quantization is the priced registration of a bit. Magnetism carries a bit, and it has always been quantized: magnetic flux through a superconducting ring comes in units of $h/2e$ (Deaver and Fairbank; Doll and Näbauer, 1961), and Dirac showed a single monopole would force every charge to be quantized (1931). Gravity, one-signed, has no off-locus point and so no bit to register; its long resistance to quantization is, on this reading, a fact of type and not of difficulty. This paper writes no Lagrangian, no propagator and no unification of couplings; it states why one branch was always going to quantize and the other was not, and leaves the field theory refused.

11. 11. FALSIFIERS AND OWED DEEDS

F-Monopole: an isolated magnetic charge, breaking the pair law of Section 6. **F-Mass:** a negative gravitational mass (the possibility examined by Bondi, 1957), creating an off-locus point and a bit for gravity. **F-Loop:** a gravitational hysteresis, a cycle of configurations returning with a remanent field. Nothing is owed on the rows' channels; one question sharpens the typing without moving it: whether gravitational radiation is a record on the reversible branch or a priced leak.

Ledger status. The barrier and the fold move from kernel theorems to WITNESSED in the magnet and the orbit; the least-erasure posit's shape is WITNESSED in the fold; the Poincaré row stays CROSSED at full grade and is now WITNESSED from the arrow side; the dot stands, and what it owes is a seat (`dot_iff_seat_absent`); the posit's shape is the fold by theorem (`posit_is_the_fold`).

12. 12. METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author's verification program, whose vocabulary is confined to this section. Its tokens: $[\perp]$ a field sealed, $[\perp T]$ a shape sealed at theorem grade, $[\Xi_0]$ a formal string held in suspension by an orientation-blind reader on a line, $[.]$ the dot. Grades join at the weakest link: a kernel theorem, a cited measurement or computation joined to a kernel schema, and a posit beside the Root Axiom, the program's one declared premise, that to exist is to actuate. Every claim here is at corroboration grade, a constructed or measured instance joined to a template theorem, and promotes nothing; $\Delta M = 0$ against the floor, mathematics being inherited and never authored.

The seventh. In the author's register the four parts of the Riemann closure are three plus one, and nothing escapes them; the six witnesses are six days, and the ledger's closure is the seventh, the Rest (Istawa), where the statuses are computed and what each row owes is one object of a computed kind. In the Lean codex's own tokens a crossed row renders $[\perp \blacklozenge \cdot \text{kinetic}]$ or $[\perp \blacklozenge \cdot \text{formal}]$, a pending row renders $[\blacklozenge_0 \times 1]$ on its channel, one owed slot and one bit from sealed (`one_bit_apart`), and the dot stands as $[.]$.

Disclosure. The author declares the reading of each correspondence as his own act; the measurements and prior theorems are their authors', cited by name; the kernel theorems are printed with their dependencies in Appendix A; every figure is computed in the run printed in Appendix B.

Provenance. Development provenance is the paper trail: the Riemann closure, Nothing Escapes, Three Plus One (10.5281/zenodo.22976494); the twenty-three-row closure, Nothing Escapes, Twenty-Three Rows (10.5281/zenodo.22985873); the companion witnesses The Electron Is the Seat, The Neutrino Is the Witness, The Arrow Has Two Branches, The Proton Is the Lock, The Magnet Is the Witness of the Pair Correlation and The Fluid Is the Witness; the Unabridged Trisduction Codex and the Lean Codex on the public register (github.com/1000sapients/Trisduction); and the author's Structural Physics register. This edition is forged and awaits a joint adversarial audit with its companions.

References. P. A. M. Dirac, Proc. R. Soc. A 133 (1931) 60. E. C. Stoner and E. P. Wohlfarth, Phil. Trans. R. Soc. A 240 (1948) 599. H. Bondi, Rev. Mod. Phys. 29 (1957) 423. R. Landauer, IBM J. Res. Dev. 5 (1961) 183. B. S. Deaver and W. M. Fairbank, Phys. Rev. Lett. 7 (1961) 43. R. Doll and M. Näbauer, Phys. Rev. Lett. 7 (1961) 51. L. Verlet, Phys. Rev. 159 (1967) 98. D. Friedan, Phys. Rev. Lett. 45 (1980) 1057. G. Perelman, arXiv:math/0211159 and math/0303109. A. Béruit et al., Nature 483 (2012) 187. J. Hong et al., Sci. Adv. 2 (2016) e1501492. E. K. Anderson et al. (ALPHA-g), Nature 621 (2023) 716. M. F. Islam, Nothing Escapes, Three Plus One, 10.5281/zenodo.22976494. M. F. Islam, Nothing Escapes, Twenty-Three Rows, 10.5281/zenodo.22985873.

13. APPENDIX A: SPHYS_TWO_ARROWS.LEAN

SHA-256 41a0d6e356c44a55bfb22f1e21020e8a661389d02280990c6bca71f57ca90f1f, core Lean 4.19.0, standalone, no library. Exit 0. 8 theorems; the compiler's own transcript, verbatim:

```

1 'SPHYS.Arrows.gravity_no_off_locus' does not depend on any axioms
2 'SPHYS.Arrows.no_bit_without_off_locus' does not depend on any axioms
3 'SPHYS.Arrows.magnetism_pair' depends on axioms: [propext, Classical.choice, Quot.sound]
4 'SPHYS.Arrows.two_worlds_at_zero_field' does not depend on any axioms
5 'SPHYS.Arrows.branches_differ' does not depend on any axioms
6 'SPHYS.Arrows.magnetic_odd' does not depend on any axioms
7 'SPHYS.Arrows.no_reading_of_E_returns_B' depends on axioms: [propext, Quot.sound]
8 'SPHYS.Arrows.one_sign_no_remanence' does not depend on any axioms

1 /-
2 SPHYS_Two_Arrows.lean · Magnetism and gravity as the two branches of one registration ·
3 core Lean 4, no library. A one-signed field has no off-locus point, so no two worlds over a
4 record and no bit to register: the reversible branch, the retraction itself. A two-signed
5 field has pairs, remanence, two worlds over one record, the priced branch. Under time
6 reversal the electric record is even and the magnetic target is odd, so no function of the
7 record returns the target. Nothing here is a field theory; it is the typing of two arrows.
8 -/
9 namespace SPHYS.Arrows
10
11 /- Sources of one sign: every source is its own conjugate, so none is off the locus. -/
12 def gravSource := { m : Int // 0 < m }
13 def conjG (s : gravSource) : gravSource := s
14 theorem gravity_no_off_locus (s : gravSource) : conjG s = s := rfl
15
16 /- A row with every instance compliant admits no erasure and hence no two worlds: there is
17 no bit for a record to miss. This is why the reversible branch keeps no record. -/
18 def Value (S : Type) (P : S → Prop) (Z : S → Prop) : Prop := ∀ s, Z s → P s
19 def Erases (S : Type) (P : S → Prop) (Z : S → Prop) : Prop := ∃ s, Z s ∧ ¬ P s
20 theorem no_bit_without_off_locus {S : Type} (P : S → Prop) (hall : ∀ s, P s) (Z : S → Prop) :
21   Value P Z ∧ ¬ Erases P Z :=
22   (fun s _ => hall s, fun (s, _) hp => hp (hall s))
23
24 /- Charges of two signs come in pairs under conjugation: an off-locus point has a partner,
25 distinct and off the locus. No monopole: the pair never separates. -/
26 def charge := Int
27 def conjC (q : Int) : Int := -q
28 theorem magnetism_pair (q : Int) (hq : q ≠ 0) : conjC q ≠ q ∧ conjC q ≠ 0 := by
29   unfold conjC; omega
30
31 /- Remanence: at zero applied field the material sits in one of two states with the same
32 record. Two worlds over one record, the kept bit. -/
33 abbrev MagState := Int × Bool -- (applied field H, remanent orientation)
34 def record (s : MagState) : Int := s.1
35 theorem two_worlds_at_zero_field :
36   record (0, true) = record (0, false) ∧ (0, true) ≠ ((0, false) : MagState) := by decide
37
38 /- The priced and the reversible branch: a cycle of the priced branch pays at least one
39 floor; the reversible branch pays none. -/
40 def cost : Bool → Nat | true => 1 | false => 0 -- true = irreversible (priced)
41 theorem priced_pays : cost true = 1 := rfl
42 theorem reversible_free : cost false = 0 := rfl
43 theorem branches_differ : cost true ≠ cost false := by decide
44
45 /- Time reversal on the field record: the electric part is even, the magnetic part odd.
46 No function of the even part returns the odd part. -/
47 abbrev Field := Int × Int -- (E, B)
48 def timeRev (f : Field) : Field := (f.1, -f.2)
49 theorem electric_even (f : Field) : (timeRev f).1 = f.1 := rfl
50 theorem magnetic_odd (f : Field) : (timeRev f).2 = -f.2 := rfl
51 theorem no_reading_of_E_returns_B (g : Int → Int) : ¬ (∀ f : Field, g f.1 = f.2) := by
52   intro h
53   have h1 := h (0, 1); have h2 := h (0, -1)
54   simp at h1 h2; omega
55
56 /- A forbidden observation, derived: remanence needs two worlds over one record, so a field whose
57 every instance is compliant can show no hysteresis, whatever its record. -/
58 theorem one_sign_no_remanence {S R : Type} (P : S → Prop) (record : S → R) (hall : ∀ s, P s) :
59   ¬ ∃ s1 s2, record s1 = record s2 ∧ P s1 ∧ ¬ P s2 :=
60   fun (s1 s2, h1 h2) => hn (hall s2)
61
62 end SPHYS.Arrows
63 #print axioms SPHYS.Arrows.gravity_no_off_locus
64 #print axioms SPHYS.Arrows.no_bit_without_off_locus
65 #print axioms SPHYS.Arrows.magnetism_pair
66 #print axioms SPHYS.Arrows.two_worlds_at_zero_field
67 #print axioms SPHYS.Arrows.branches_differ
68 #print axioms SPHYS.Arrows.magnetic_odd
69 #print axioms SPHYS.Arrows.no_reading_of_E_returns_B
70 #print axioms SPHYS.Arrows.one_sign_no_remanence

```

14. APPENDIX B: ARROWS_TWIN.F90 AND ITS RUN

SHA-256 4cf40ecbb1aa8341f0c32b70811ca31f53cf6ceaaba3d1e00e6d173623ea6641, standard Fortran 2018, zero warnings under gfortran 13.3.0 -std=f2018 -O2 -fno-fast-math -ffp-contract=off -Wall -Wextra. Nine checks, zero failures; every figure computed in this run.

```

1 ! Arrows_Twin.f90 · twin of "The Arrow Has Two Branches" · Fortran 2018.
2 ! The priced branch: a Stoner-Wohlfarth nanomagnet swept through its hysteresis loop. The
3 ! reversible branch: a Kepler orbit integrated by velocity Verlet and run backwards. Every
4 ! figure computed in this run; cited constants labelled; an oracle stops on any failure.
5 program arrows_twin
6   implicit none
7   integer, parameter :: dp = selected_real_kind(15, 307)
8   real(dp), parameter :: PI = 3.14159265358979323846_dp, KB = 1.380649e-23_dp
9   real(dp) :: psi, h, th, m_desc0, m_asc0, area, hsw_calc, hsw_theory, mprev, hprev
10  real(dp) :: x(2), v(2), a(2), x0(2), v0(2), e0, e1, dt, work, r, fx(2), xold(2)
11  real(dp) :: kt, floor, per_switch, emax_drift, ke0, kel
12  integer :: i, nsteps, nchk, nfail, k
13  logical :: switched
14  nchk = 0; nfail = 0
15  ! ----- the priced branch: e(th) = sin^2(th)/2 - h cos(th - psi), field along psi
16  psi = 5.0_dp*PI/180.0_dp; th = 0.0_dp; area = 0.0_dp; switched = .false.; hsw_calc = 0.0_dp
17  mprev = cos(th - psi); hprev = 2.0_dp
18  do i = 0, 4000
19     h = 2.0_dp - 4.0_dp*real(i,dp)/4000.0_dp

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20  call relax(th, h, psi)
21  if (abs(h) < 1.0e-12_dp) m_desc0 = cos(th - psi)
22  if (.not. switched .and. cos(th - psi) < 0.0_dp) then
23      switched = .true.; hsw_calc = -h
24  end if
25  area = area + 0.5_dp*(cos(th - psi) + mprev)*(h - hprev)
26  mprev = cos(th - psi); hprev = h
27  end do
28  do i = 0, 4000                                ! ascending branch, h from -2 to 2
29      h = -2.0_dp + 4.0_dp*real(i,dp)/4000.0_dp
30      call relax(th, h, psi)
31      if (abs(h) < 1.0e-12_dp) m_asc0 = cos(th - psi)
32      area = area + 0.5_dp*(cos(th - psi) + mprev)*(h - hprev)
33      mprev = cos(th - psi); hprev = h
34  end do
35  area = abs(area)
36  hsw_theory = (cos(psi)**(2.0_dp/3.0_dp) + sin(psi)**(2.0_dp/3.0_dp))**(-1.5_dp)
37  call check('two worlds at zero field: the descending branch holds +m', m_desc0 > 0.9_dp)
38  call check('two worlds at zero field: the ascending branch holds -m', m_asc0 < -0.9_dp)
39  call check('switching field matches the Stoner-Wohlfarth astroid to 1 percent', abs(hsw_calc/hsw_theory - 1.0_dp) < 1.0e-2_dp)
40  call check('the loop encloses area: the cycle is priced', area > 1.0_dp)
41  kt = KB*300.0_dp; floor = kt*log(2.0_dp)
42  per_switch = area * 40.0_dp * kt                                ! work per cycle = area x 2KV; per switch = area x KV; KV = 40 kT
43  call check('a thermally stable bit (KV = 40 kT) pays far above the Landauer floor per switch', per_switch > 10.0_dp*floor)
44  call check('the Landauer floor at 300 K is 2.87e-21 J', abs(floor/2.87e-21_dp - 1.0_dp) < 2.0e-3_dp)
45  ! ----- the reversible branch: Kepler orbit, GM = 1, velocity Verlet
46  x0 = [1.0_dp, 0.0_dp]; v0 = [0.0_dp, 1.2_dp]; x = x0; v = v0; dt = 1.0e-3_dp
47  e0 = 0.5_dp*dot_product(v,v) - 1.0_dp/norm2(x); a = -x/norm2(x)**3
48  nsteps = 150000; emax_drift = 0.0_dp; work = 0.0_dp; ke0 = 0.5_dp*dot_product(v,v)
49  do k = 1, nsteps
50      xold = x
51      x = x + v*dt + 0.5_dp*a*dt*dt
52      fx = -x/norm2(x)**3
53      work = work + 0.5_dp*dot_product(a + fx, x - xold)
54      v = v + 0.5_dp*(a + fx)*dt; a = fx
55      e1 = 0.5_dp*dot_product(v,v) - 1.0_dp/norm2(x)
56      emax_drift = max(emax_drift, abs(e1/e0 - 1.0_dp))
57  end do
58  ke1 = 0.5_dp*dot_product(v,v)
59  call check('energy returns: relative drift over ten orbits below 1e-6', emax_drift < 1.0e-6_dp)
60  v = -v                                ! reverse the arrow of the orbit
61  do k = 1, nsteps
62      x = x + v*dt + 0.5_dp*a*dt*dt
63      fx = -x/norm2(x)**3
64      v = v + 0.5_dp*(a + fx)*dt; a = fx
65  end do
66  r = norm2(x - x0)
67  call check('run backwards, the orbit retraces to its start within 1e-8', r < 1.0e-8_dp)
68  call check('work-energy: the path integral of gravity equals the change in kinetic energy', abs(work - (ke1 - ke0)) < 1.0e-5_dp)
69  write(*, '(a,2f10.5)') ' remanent m at h = 0 (desc, asc) = ', m_desc0, m_asc0
70  write(*, '(a,2f10.5)') ' switching field (computed, SW) = ', hsw_calc, hsw_theory
71  write(*, '(a,f10.5)') ' loop area (units mu0 Ms HK) = ', area
72  write(*, '(a,es11.4,a)') ' Landauer floor kT ln 2 at 300 K = ', floor, ' J'
73  write(*, '(a,es11.4,a)') ' dissipation per switch, KV = 40kT = ', per_switch, ' J'
74  write(*, '(a,es11.4)') ' Kepler energy drift (max, rel) = ', emax_drift
75  write(*, '(a,es11.4)') ' Kepler return error after reversal= ', r
76  write(*, '(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', nchk, ', "failures":', nfail, '}'
77  if (nfail > 0) error stop 1
78  contains
79  subroutine relax(th, h, psi)
80      real(dp), intent(inout) :: th
81      real(dp), intent(in) :: h, psi
82      integer :: it
83      real(dp) :: g
84      do it = 1, 20000
85          g = sin(th)*cos(th) + h*sin(th - psi)
86          th = th - 0.05_dp*g
87          if (abs(g) < 1.0e-13_dp) exit
88      end do
89  end subroutine relax
90  subroutine check(label, cond)
91      character(*), intent(in) :: label
92      logical, intent(in) :: cond
93      nchk = nchk + 1
94      if (.not. cond) then
95          nfail = nfail + 1; write(*, '(a,a)') ' FAIL ', label
96      end if
97  end subroutine check
98  end program arrows_twin

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The run, verbatim.

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1  remanent m at h = 0 (desc, asc) = 0.99619 -0.99619
2  switching field (computed, SW) = 0.76700 0.76643
3  loop area (units mu0 Ms HK) = 3.03593
4  Landauer floor kT ln 2 at 300 K = 2.8710E-21 J
5  dissipation per switch, KV = 40kT = 5.0299E-19 J
6  Kepler energy drift (max, rel) = 2.8215E-07
7  Kepler return error after reversal= 1.5120E-12
8  BATTERY-JSON: {"checks":9,"failures":0}

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